

MAP 2202 - Real Analysis II

Assignment # 02

ASP/19/20/049 - 4809 - B.L.P. Nishani

89%

- ① Find the interval of convergence of each of the following power series.

(i) $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$

80

$$\lambda = \limsup |a_n|^{1/n} = \limsup \left| \frac{1}{2^n} \right|^{1/n} \text{ as } a_n = \frac{1}{2^n} \forall n \in \mathbb{N}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2}$$

$\therefore$ The radius of convergence,

$$R = \frac{1}{\lambda} = \frac{1}{1/2} = 2 //$$

The power series converges absolutely if $|x| < 2$.
i.e. $-2 < x < 2$

When $x = 2$ the power series $\sum_{n=0}^{\infty} \frac{2^n}{2^n} = \sum_{n=0}^{\infty} 1^n$ is divergent.

When $x = -2$

$$\sum_{n=0}^{\infty} \frac{x^n}{2^n} = \sum_{n=0}^{\infty} \frac{-2^n}{2^n} = \sum_{n=0}^{\infty} (-1)^n \text{ is divergent.}$$

using test show them.

$\therefore$ The interval of convergence

$$IC = (-2, 2)$$

No: _____

Date: ____/____/____

(iv) $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{(k+1)!} x^{2n}$

Here $a_n = \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{(k+1)!} x^{2n}$

Here $a_{n+1} = \frac{1 \cdot 3 \cdot \dots \cdot (2n+1)}{(k+1)!}$

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{(k+1)!} \times \frac{(k+1)!}{1 \cdot 3 \cdot \dots \cdot (2n+1)} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{1}{(2n+1)} \right|$$

$$R = 0 < 1$$

$\therefore$ The power series converges only at $x = 0$ (center)

② $\sum_{n=0}^{\infty} \frac{(n+1)}{(n+2)(n+3)} x^n$

70

X

Here

$a_n = \frac{(n+1)}{(n+2)(n+3)}$

$a_{n+1} = \frac{(n+2)}{(n+3)(n+4)}$

$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+2)}{(n+3)(n+4)} \cdot \frac{(n+2)(n+3)}{(n+1)} = \lim_{n \rightarrow \infty} \frac{(n+2)^2}{(n+1)(n+4)} = 1$

$a_{n+1} = \frac{(n+2)}{(n+3)(n+4)}$

$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+2)}{(n+3)(n+4)} \cdot \frac{(n+2)(n+3)}{(n+1)} = \lim_{n \rightarrow \infty} \frac{(n+2)^2}{(n+1)(n+4)} = 1$

$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$

$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)}{(n+2)(n+3)} \cdot \frac{(n+3)(n+4)}{(n+2)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

✓

Radius of convergence $R = 1$

∴ The power series converges absolutely if $|x| < 1$

$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(n+4)}{(n+2)^2} \right| = 1$

When $x = 1$

$$\sum_{n=0}^{\infty} \frac{C_{n+1}}{(n+2)(n+3)} (1)^n = \sum_{n=0}^{\infty} \frac{C_{n+1}}{(n+2)(n+3)}$$

is convergent. ~~X~~

When $x = -1$

$$\sum_{n=0}^{\infty} \frac{C_{n+1}}{(n+2)(n+3)} (-1)^n$$

is convergent. ✓

 $\therefore$ The interval of convergence

$$IC = [-1, 1]$$

~~X~~

③ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot x^n$

70

Here $a_n = \frac{(-1)^{n+1}}{n}$

$$a_{n+1} = \frac{(-1)^{n+2}}{(n+1)}$$

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{n} \cdot \frac{(n+1)}{(-1)^{n+2}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n+1}{-n} \right|$$

No: _____

Date: ③ / /

$$= \lim_{n \rightarrow \infty} \frac{n+1}{n}$$

$$= \lim_{n \rightarrow \infty} 1 + \frac{1}{n}$$

$$= 1 //$$

Radius of convergence $R = 1$ $\therefore$ The power series converges absolutely if $|x| < 1$

$$\text{i.e. } -1 < x < 1$$

When $x = 1$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} 1^n = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is divergent. $\times$

When $x = -1$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (-1)^n = \sum_{n=1}^{\infty} \frac{(-1)}{n} \text{ is divergent}$$

 $\therefore$ The interval of convergence

$$\text{IC} = (-1, 1)$$

This is the closed form of the power series

(04) (i) $\sum_{n=1}^{\infty} 9x^{3n-1}$

$$S = \frac{a}{1-r}$$

$$a = 9x^2$$

$$r = x^3$$

$$\therefore S = \frac{9x^2}{1-x^3}$$

This is valid for all $|x^3| < 1$, i.e. $-1 < x < 1$.
 $\therefore$ The open interval is $(-1, 1)$.

(iv) $\sum_{n=0}^{\infty} \frac{x^{3n}}{n+1}$

$$S = \frac{a}{1-r}$$

$$a = x^3/2$$

$$r = x^3$$

$$S = \frac{x^3/2}{1-x^3} = \frac{x^3}{2(1-x^3)}$$

This is the closed form of the power series.

⑥ Show that

$$\ln(x+1) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \text{ where } -1 \leq x \leq 1$$

(16c)

Deduce that

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

Let $f(x) = \ln(x+1)$, and $C_n = \frac{f^{(n)}(0)}{n!}$

$$f(x) = \ln(x+1) \Rightarrow f(0) = 0 \Rightarrow C_0 = 0$$

$$f'(x) = \frac{1}{x+1} \Rightarrow f'(0) = 1 \Rightarrow C_1 = 1$$

$$f''(x) = \frac{-1}{(1+x)^2} \Rightarrow f''(0) = -1 \Rightarrow C_2 = -1/2$$

$$f'''(x) = \frac{2}{(1+x)^3} \Rightarrow f'''(0) = 2 \Rightarrow C_3 = \frac{2}{3!} = \frac{1}{3}$$

$$f^{(4)}(x) = \frac{-6}{(1+x)^4} \Rightarrow f^{(4)}(0) = -6 \Rightarrow C_4 = \frac{-6}{4!} = -\frac{1}{4}$$

$$\therefore \ln(1+x) = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + \dots$$

$$= 0 + 1 \cdot x + (-1/2)x^2 + (1/3)x^3 - (1/4)x^4 + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \text{ Where } |x| \leq 1$$

No. _____

Date: ____/____/____

When $x=1$

$$\ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$\therefore \ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots //$$

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$\ln(x) = \ln(x+1) - \ln(x)$$

$$0 = 0 \Rightarrow 0 = \ln(x) \Rightarrow \ln(x+1) = \ln(x)$$

$$1 = 1 \Rightarrow 1 = \ln(x) \Rightarrow \ln(x+1) = \ln(x)$$

$$f'(x) = \frac{1}{x+1} \Rightarrow f''(x) = -\frac{1}{(x+1)^2}$$

$$f'''(x) = \frac{2}{(x+1)^3} \Rightarrow f^{(4)}(x) = -\frac{6}{(x+1)^4}$$

$$f^{(5)}(x) = \frac{24}{(x+1)^5}$$

$$\ln(x) = \ln(x+1) - \ln(x)$$

$$\ln(x) = \ln(x+1) - \ln(x)$$

$$\ln(x) = \ln(x+1) - \ln(x)$$